

Flight Delay Prediction Project Report

Introduction

Airline delays impact countless passengers each year, causing scheduling problems and reduced all-round efficiency. Accurately predicting flight delays can help airlines target specific problems while giving passengers with more reliable travel predictions. This project applies visualization and machine learning techniques to analyze flight data over one month and develop predictive models capable of estimating the odds of and length of departure delays.

Problem Statement

The goal of this project is to develop learning models that predict flight departure delays using a month of flight information from the Bureau of Transportation Statistics. This project identifies patterns and trends connected to flight delays, builds predictive models to estimate length of and odds of a flight delay, and evaluates each model's performance using data mining metrics.

Background

Flight delay prediction is very important for airlines; it prevents customer dissatisfaction and improves overall efficiency. An enormous number of variables are involved in flight delays, but airline carrier, departure time, flight distance, and day of the week all influence the odds in some way. Machine learning methods have been used to identify relationships between these factors to help predict delays.

The flight data analyzed in this project was obtained from the Bureau of Transportation Statistics (BTS), which maintains a colossal database of domestic airline operations in the United States. The BTS On-Time Performance dataset provides detailed information on scheduled and actual departure times, flight durations, delays, cancellations, carriers, and other relevant variables that make this sort of modeling possible.

Data Preparation and Exploratory Analysis

The project began with the collection and preprocessing of about 500,000 flight records. To improve the project's efficiency while keeping a valid representation of delays, a stratified 20% sample of approximately 100,000 observations was created that maintained the original proportion of delayed flights. Flights that were canceled or diverted were removed because they were not relevant to the prediction.

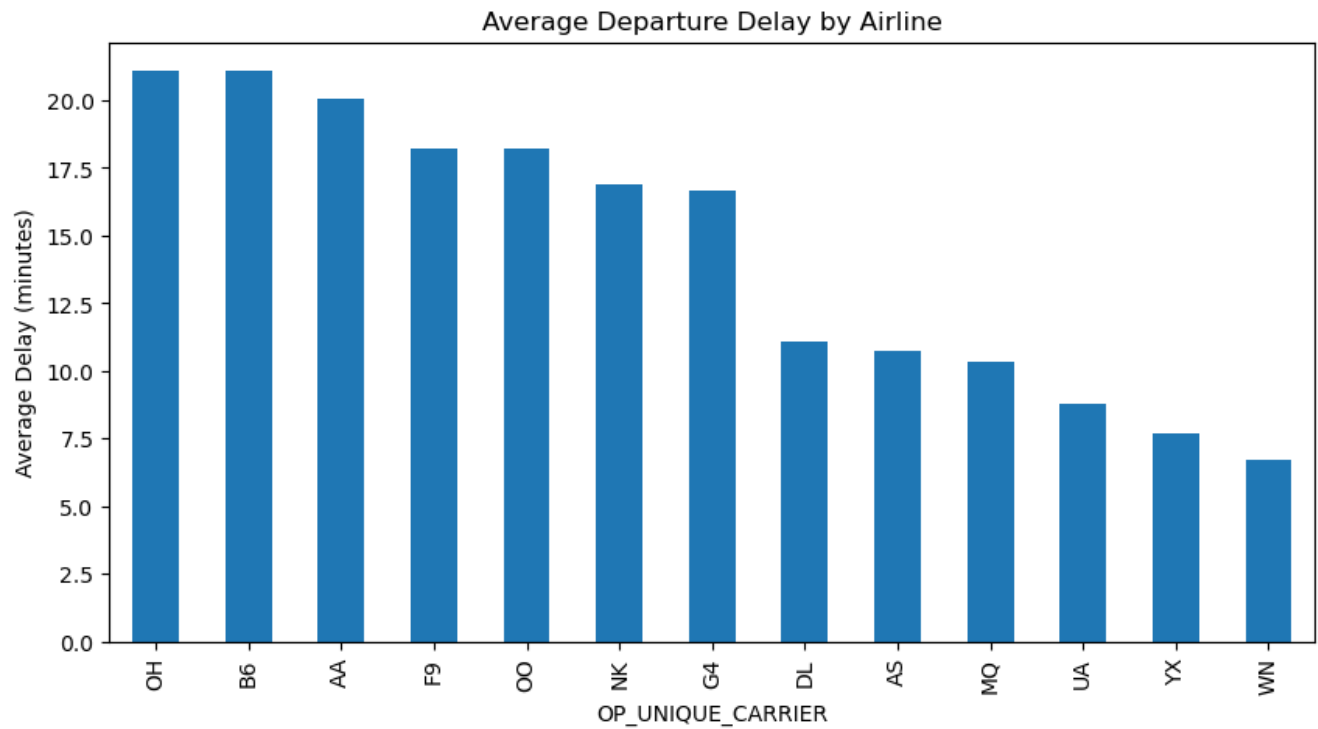
The data pipeline for MySQL was built. The database schema for the flights table was written. Then the loading script was written and moved into project.ipynb, so the team could connect to the database, load the flight dataset from the Bureau of Transportation Statistics, and bring it into a dataframe for analysis. The database password was also changed from a hardcoded value to a getpass prompt for security.

Several preprocessing steps were completed before modeling. Distance and elapsed flight time were standardized, and a departure-hour feature was constructed from the scheduled departure time.

Additional categorical variables, including airline carrier, month, and day of the week, were prepared for use in regression algorithms through one-hot encoding.

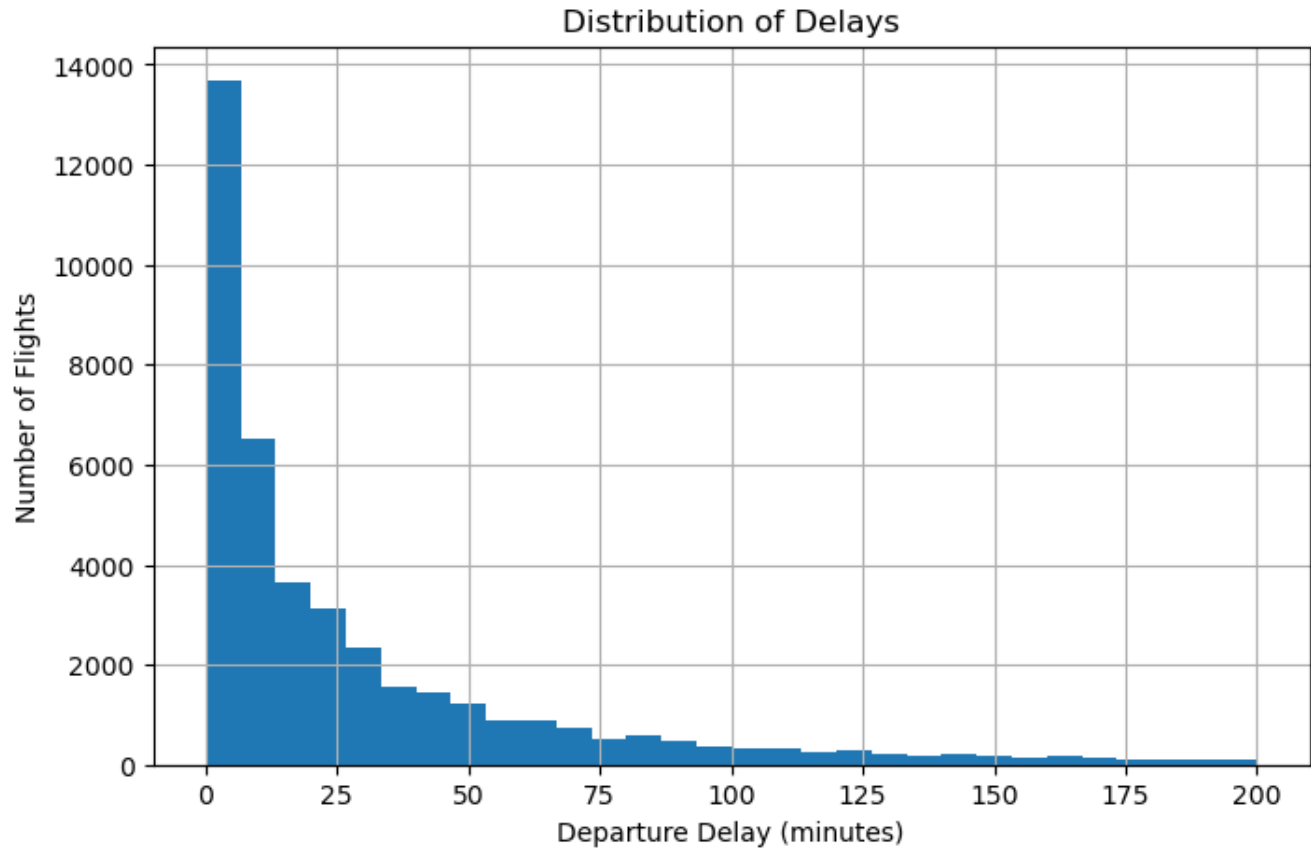
The average delay for each carrier (Fig 1.) shows that there is a significant difference across different carriers. OH (PSA Airlines) and B6 (JetBlue Airways Corporation) both had the highest average delay of over 20 minutes, though this appears to be due to different factors for each. PSA Airlines runs many short segments, causing compounded small delays for each flight throughout the day. JetBlue suffers from unrealistic scheduling and actually suffered a Department of Transportation penalty due to its chronically late flights in January 2025. WN (Southwest Airlines) performs particularly well in this dataset. This is likely due to their investment in fast turnaround times and large connected hubs.

Fig 1.



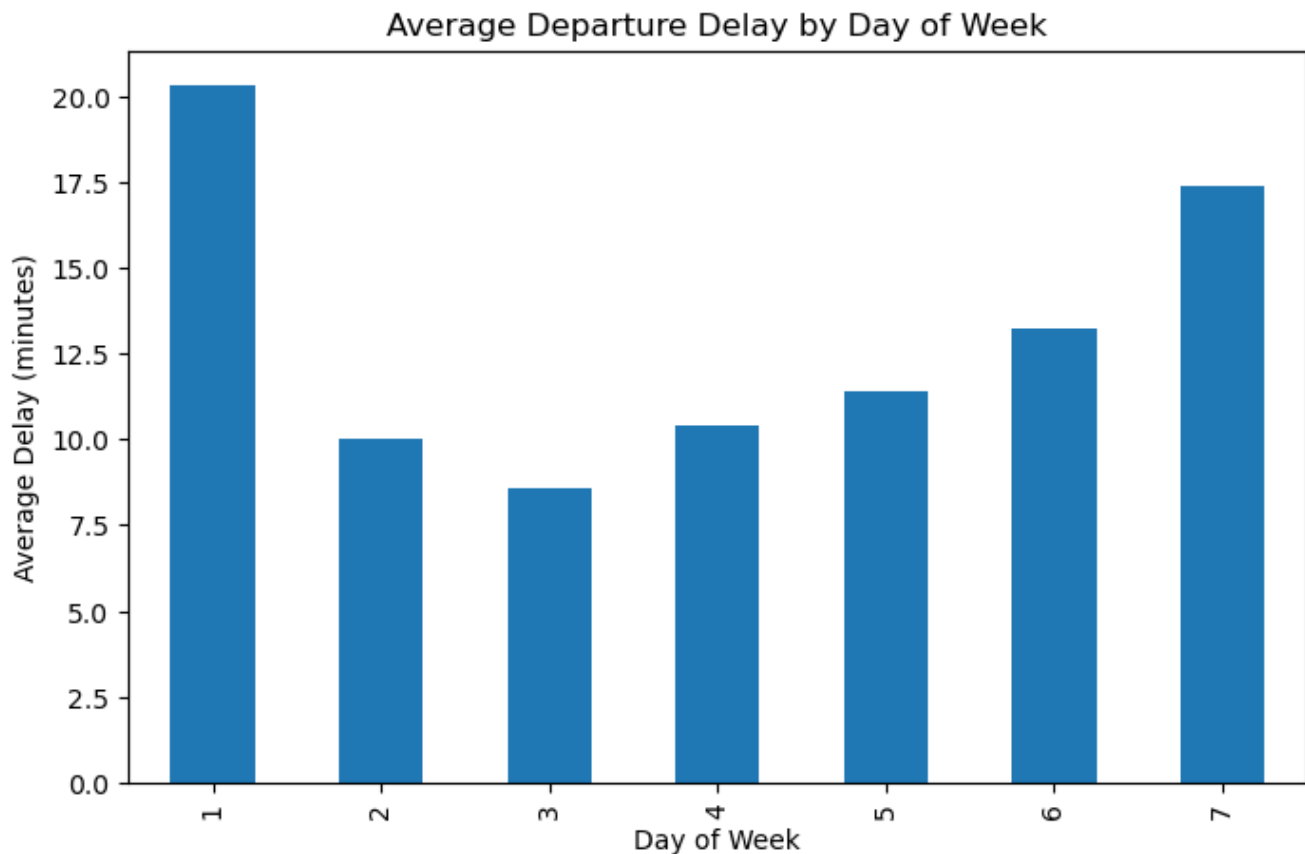
The distribution of delay magnitudes was also analyzed. Delays over 200 minutes in length and negative delay values (early arrival) are removed for improved data visualization. 0-5 minutes of delay was the most common by far, which is to be expected. Delays of 10-15 minutes are shown to be very common, with longer delays becoming rapidly less common after about 25 minutes. After 60 minutes, delay magnitude counts continue to shrink slightly as the magnitude increases. This is to be expected, as very small delays can be caused by any number of minor problems, while an hour-long delay would be caused by something substantial and out of the ordinary.

Fig 2.



Finally, average departure delay by weekday was graphed (fig 3). Monday (1) is shown to be the worst day for delays, followed by Saturday and then Sunday. Wednesday has the fewest delays. This is likely due to congestion in the start of the week for work-related travel and people returning from weekend trips. This data is particularly useful, as it can inform airlines and travelers about likely congestion periods and the best days to fly.

Fig 3.



Regression Model

A regression model that predicts the number of minutes a flight is delayed at departure was implemented. Rather than a yes/no delay classification, this model outputs a continuous prediction of delay length, giving a more granular view of expected delay severity for a given flight. The feature set used to train the model was built by combining numeric and categorical predictors. Numeric features included flight distance, scheduled elapsed flight time, and a departure-hour feature derived from the scheduled departure timestamp. Categorical features, including operating carrier, day of the week, and month, were one-hot encoded to be used by the regression algorithms. The data was split into training and test sets. Feature scaling was applied by fitting the scaler only to the training data and then applying it to the test data, to avoid leaking information from the held-out set into the training set. Instead of committing to a single algorithm up front, a comparison pipeline was built using 5-fold cross-validation on the training data to evaluate three candidate regressors: linear regression, ridge regression, and random forest regression, with performance scored by root mean squared error (RMSE). Whichever candidate performed best under cross-validation was then passed through a grid search to tune its key hyperparameters: regularization strength for ridge regression, or tree depth and count for the random forest, rather than relying on default settings. This produced a data-driven, defensible choice of final model. The selected, tuned model was evaluated once on a held-out test set it had not seen during training or model selection, using mean absolute error, mean squared error, RMSE,

and R^2 to measure how closely predicted delay minutes matched actual delays. Two supporting visualizations were also built: a feature-importance (or coefficient-magnitude) chart showing which factors most influenced the predicted delay time (Fig 4.), and a predicted-vs-actual scatterplot to visually assess model fit and identify where predictions diverged most from reality (Fig 5.). The regression pipeline was originally built and iterated on as a standalone script, then migrated to the project's shared notebook as a dedicated cell. As part of that migration, the code was restructured. Hence, it reuses an already cleaned, already sampled version of the dataset from earlier in the notebook, rather than re-reading and re-cleaning the full raw dataset (roughly 500,000 rows) a second time. This reduced the pipeline's redundant processing and kept the regression results consistent with those from the same data used elsewhere in the notebook. The work covered the full regression modeling workflow end-to-end: feature engineering, model comparison, hyperparameter tuning, held-out evaluation, and interpretability visualizations, along with integrating and optimizing the pipeline within the shared project notebook.

Fig 4.

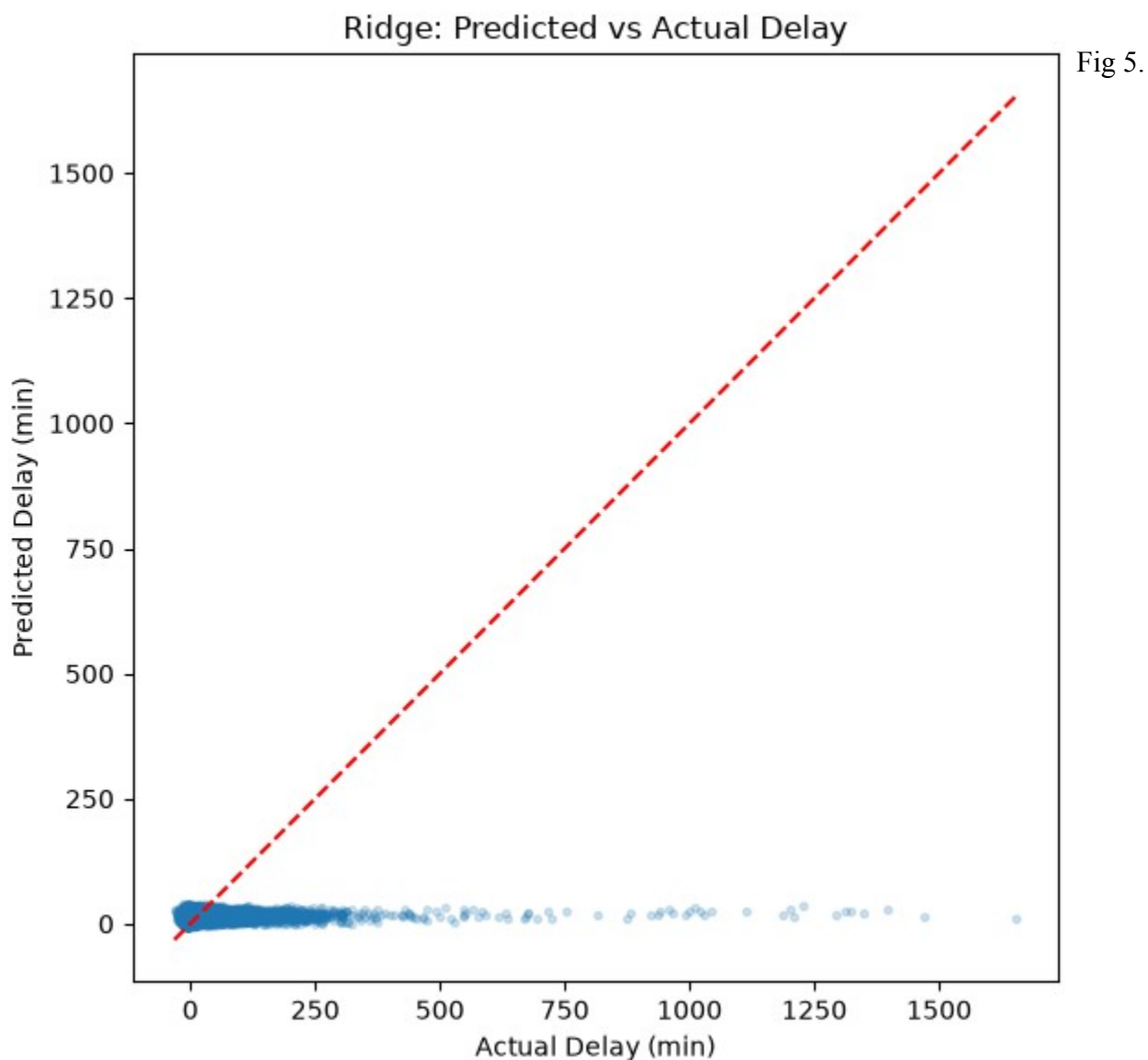
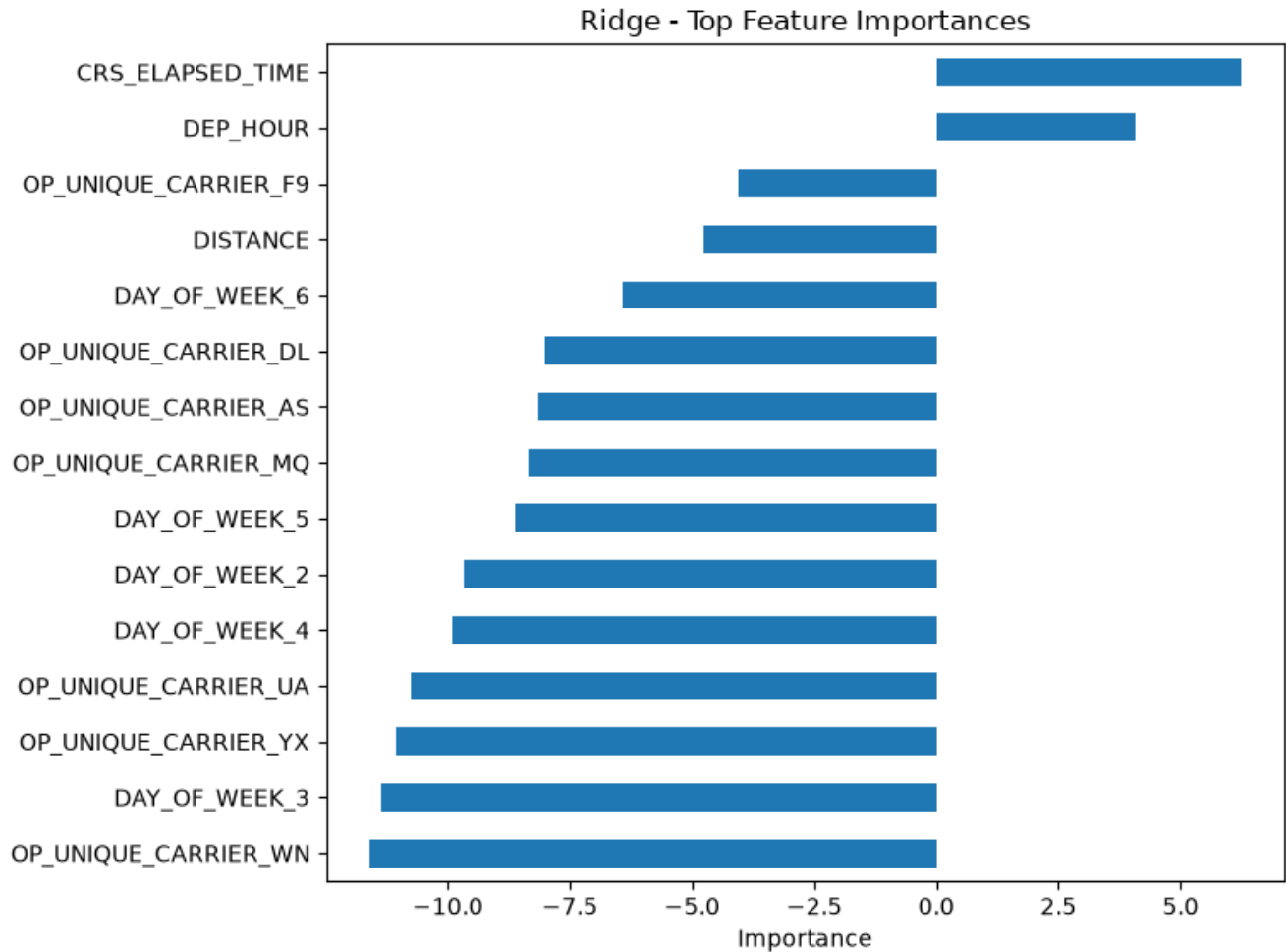


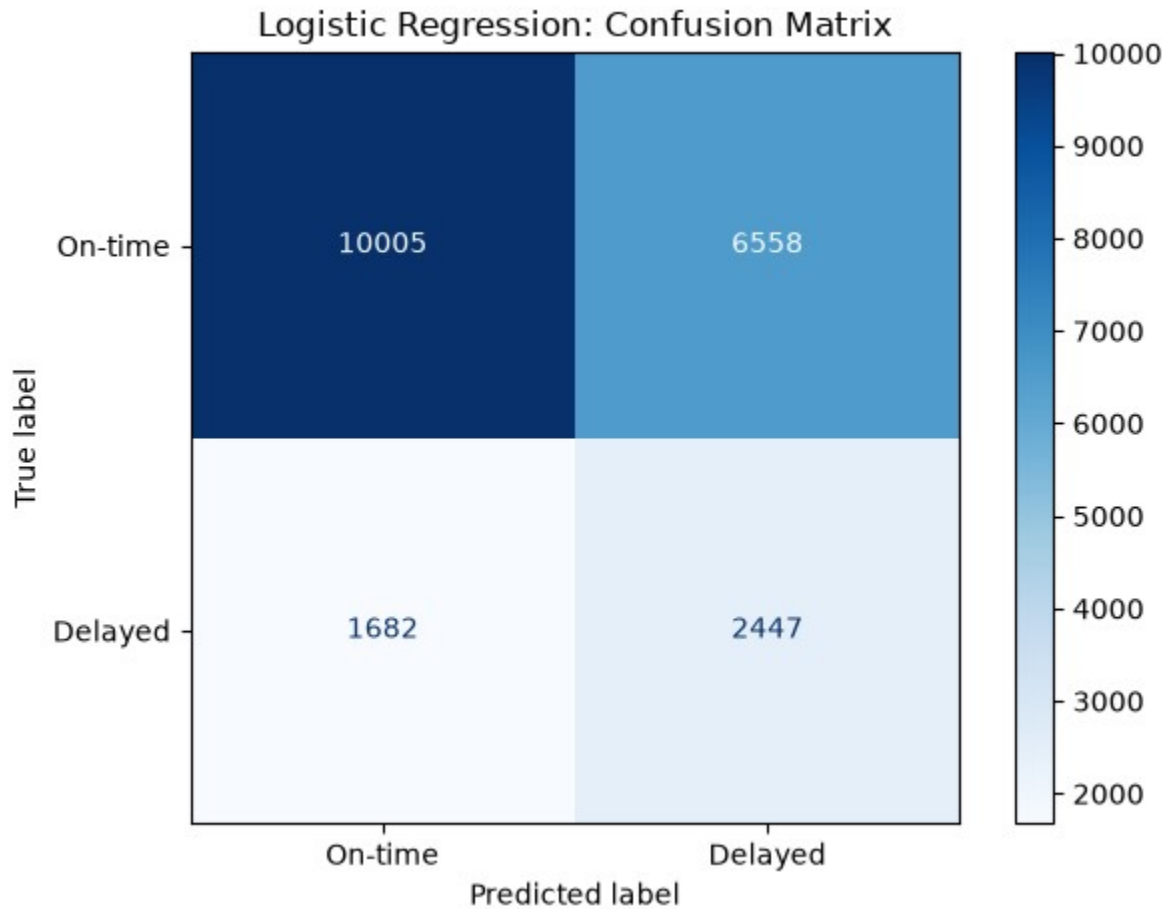
Fig 5.



Classification Model

For the modeling part, a classifier was built to predict if a flight will be delayed. Distance, scheduled elapsed time, departure hour, carrier, day of week, and month were used as features. The categorical features were encoded, and the data was split into train and test sets, keeping the same proportion of delayed flights in both sets. A logistic regression model with class weights was trained to handle the imbalance between delayed and on-time flights. (Fig 6.) It was then evaluated with accuracy, precision, recall, F1, and a confusion matrix. The model achieved 60% accuracy and identified 59% of the actual delays, but precision was low at only 27%. This means the model marks many on-time flights as delayed as well. For a system that warns passengers, this is acceptable. For a system that changes airline operations, it would not be enough.

Fig 6.



Conclusion

Flight delay prediction is complicated, and a great number of factors are involved. This project analyzes a few significant ones to identify major trends that could help travelers avoid delays by selecting the right airline or traveling on a certain day of the week. The predictive models developed could be used as the basis for something larger, helping airlines improve their services.

References

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